

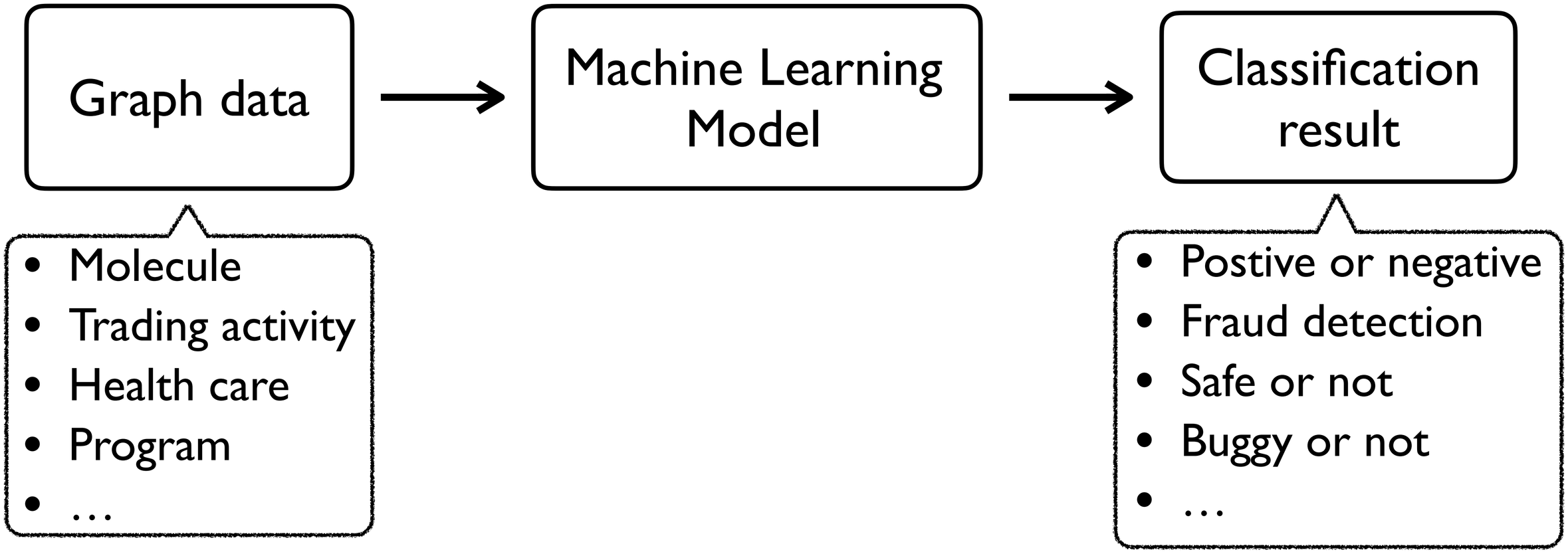
PL4XGL: A Programming Language Approach to Explainable Graph Learning

Minseok Jeon, Jihyeok Park, and Hakjoo Oh



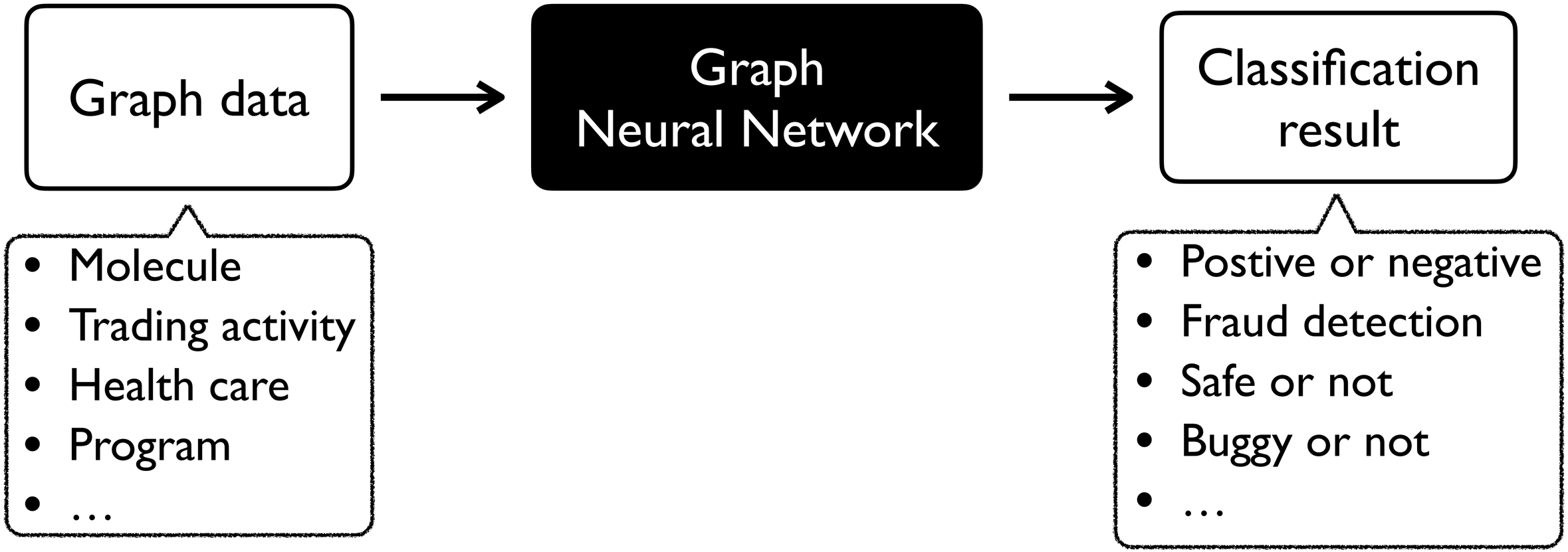
PLDI 2024 @ Copenhagen, Denmark

Graph Machine Learning



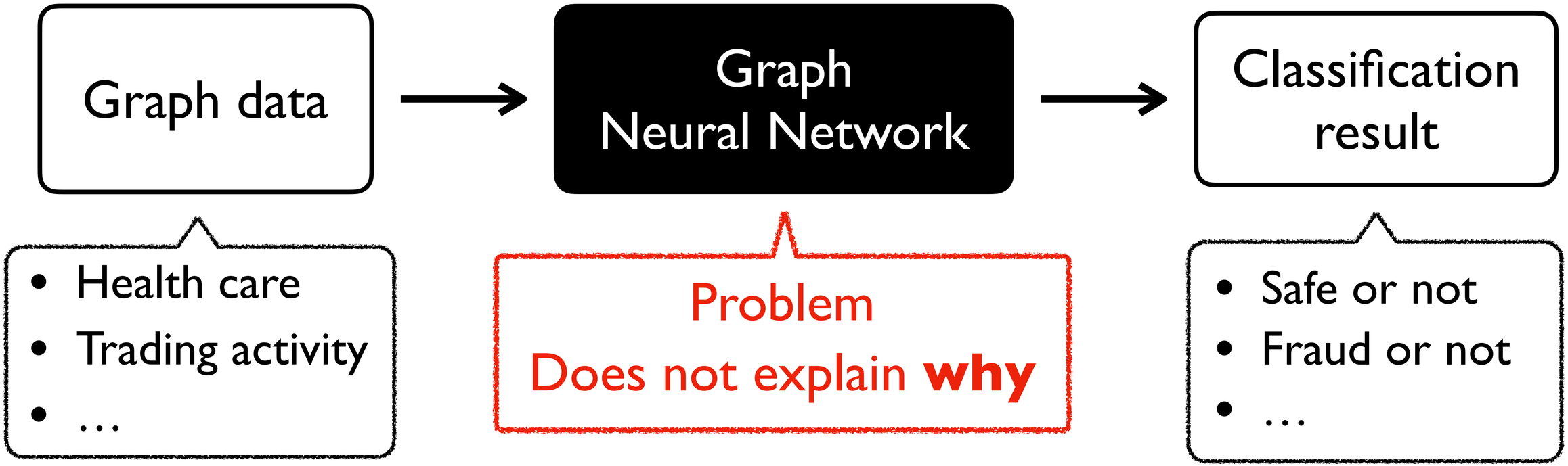
Graph Machine Learning

- Mainstream: Graph Neural Network (unexplainable AI)



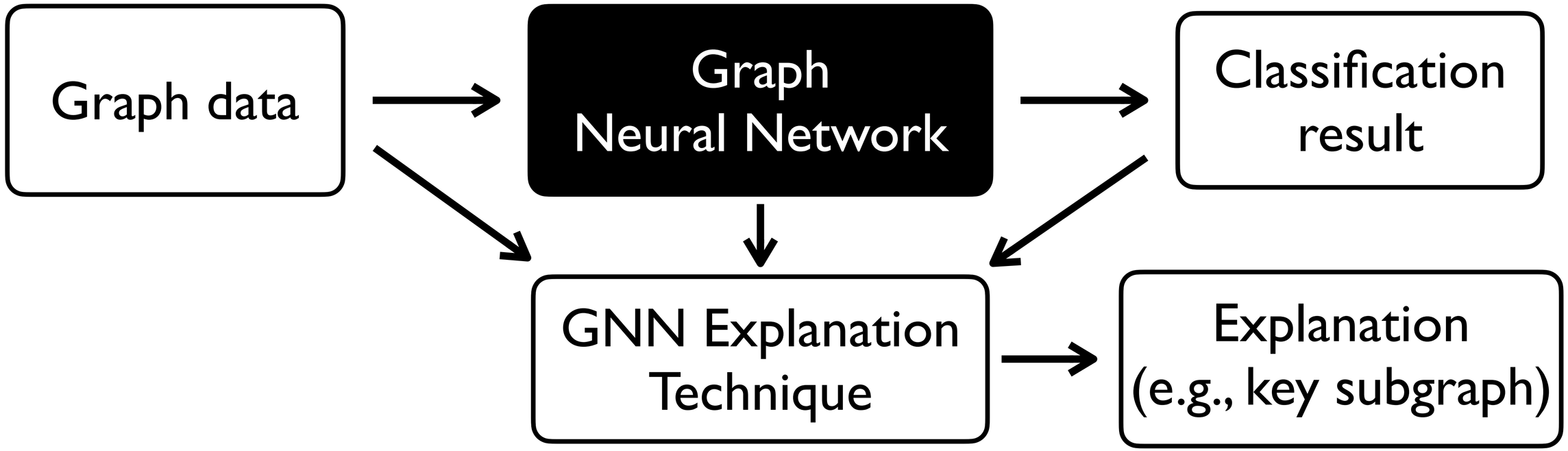
Graph Machine Learning

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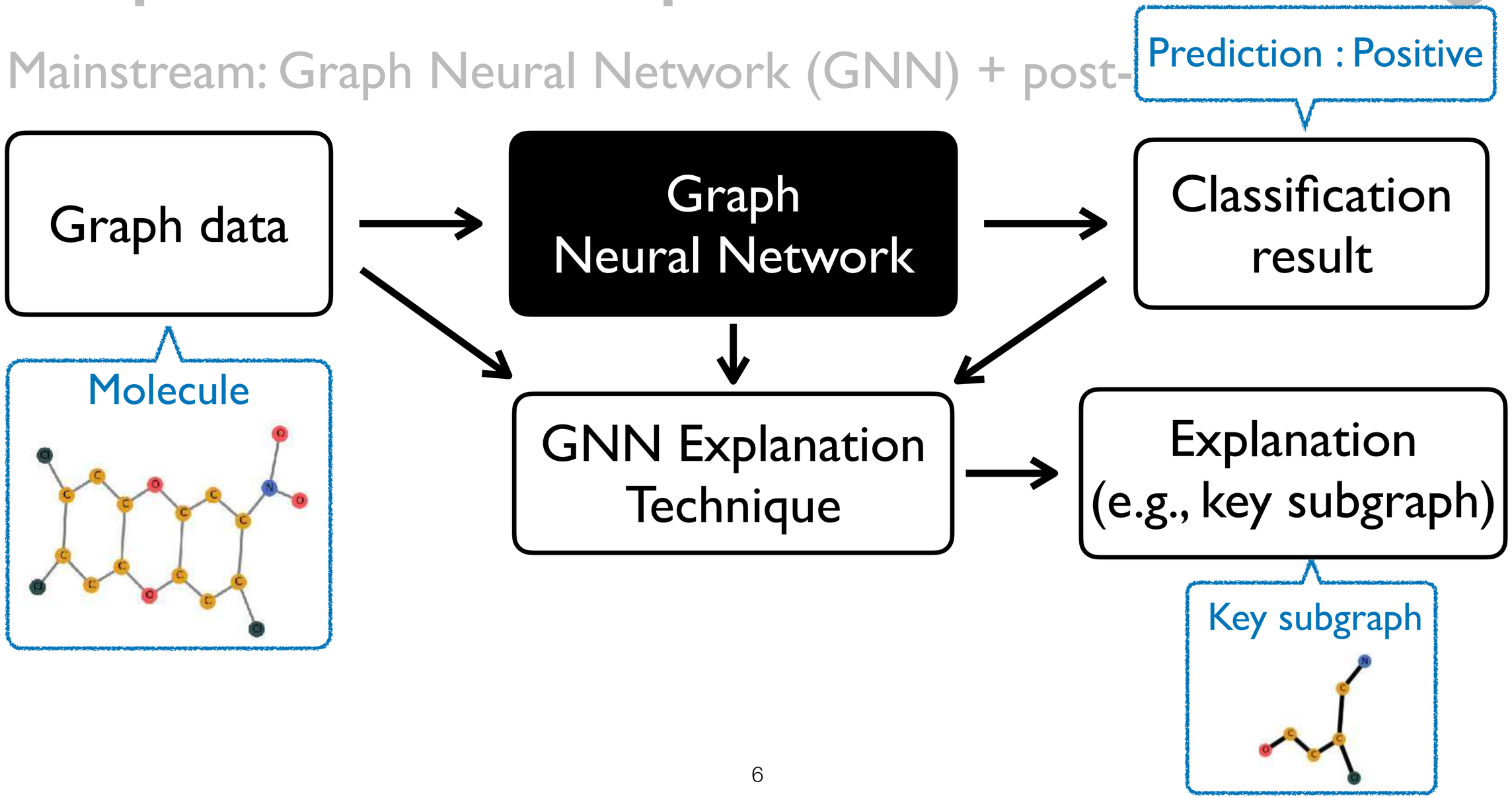
Explainable Graph Machine Learning

- Mainstream: Graph Neural Network (GNN) + post-hoc "explainers"



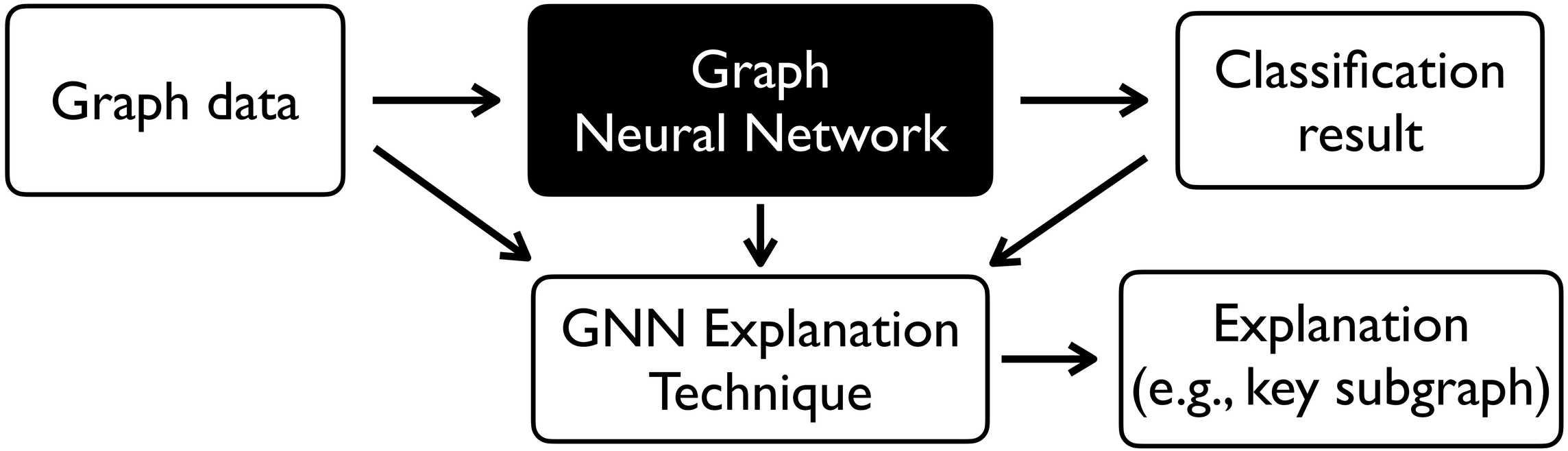
Explainable Graph Machine Learning

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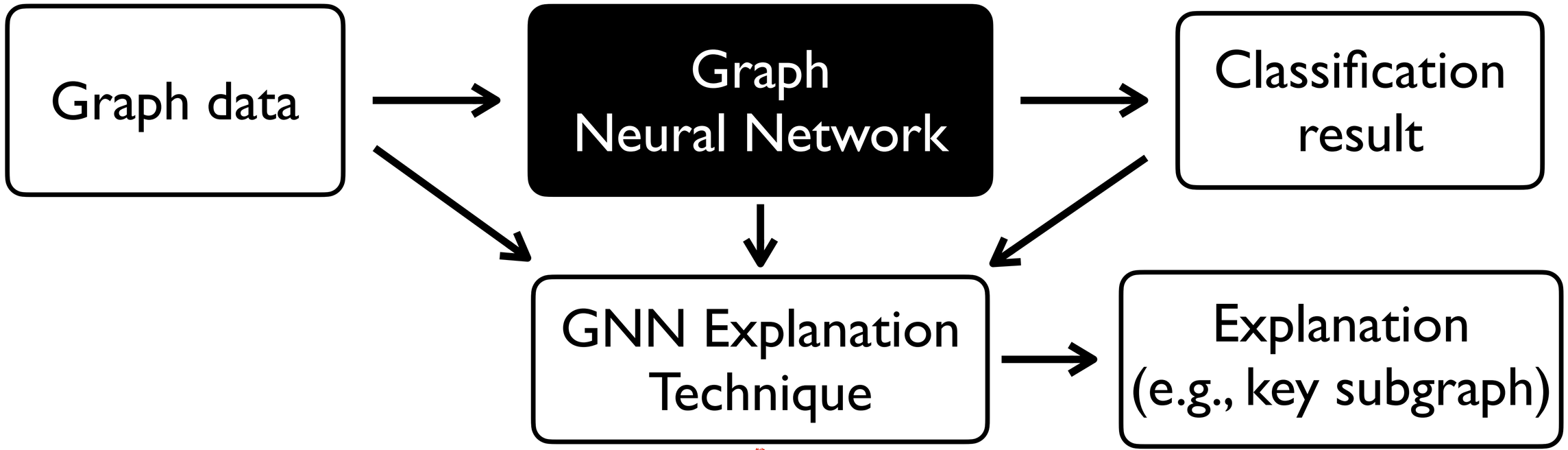
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Explainable Graph Machine Learning

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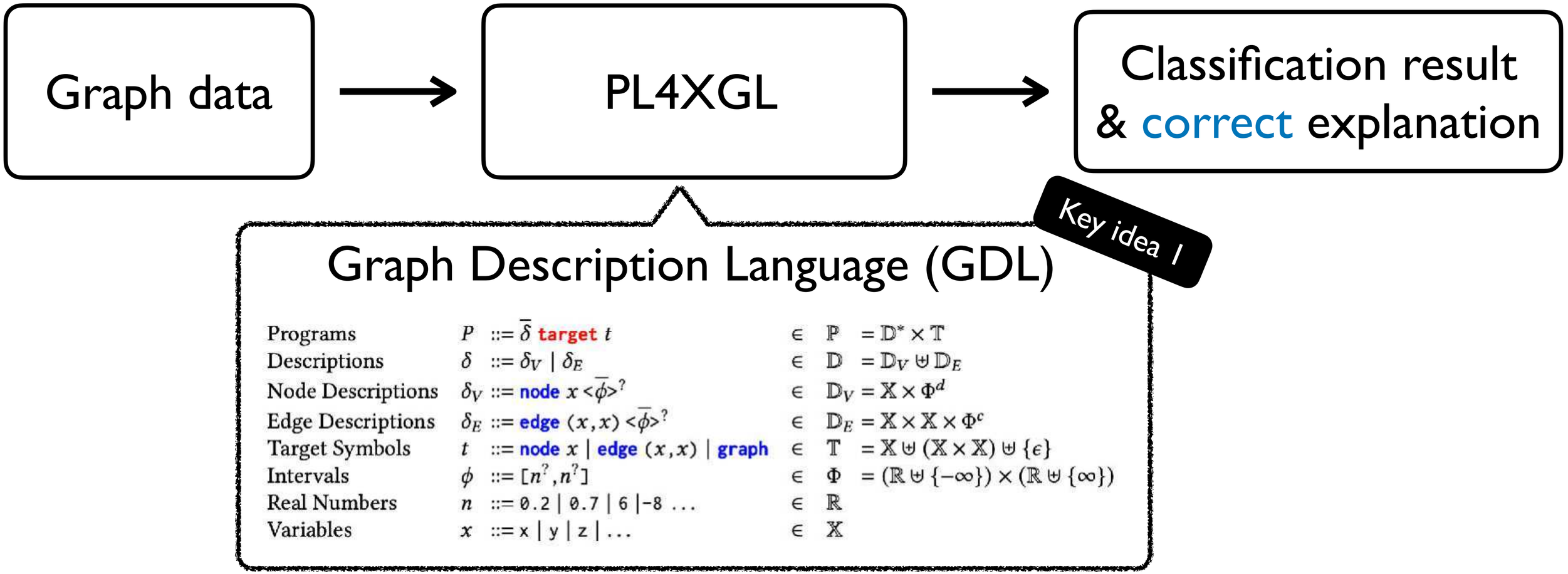


Two key limitations

- Additional (expensive) explanation cost is required
- The explanations are not guaranteed to be correct

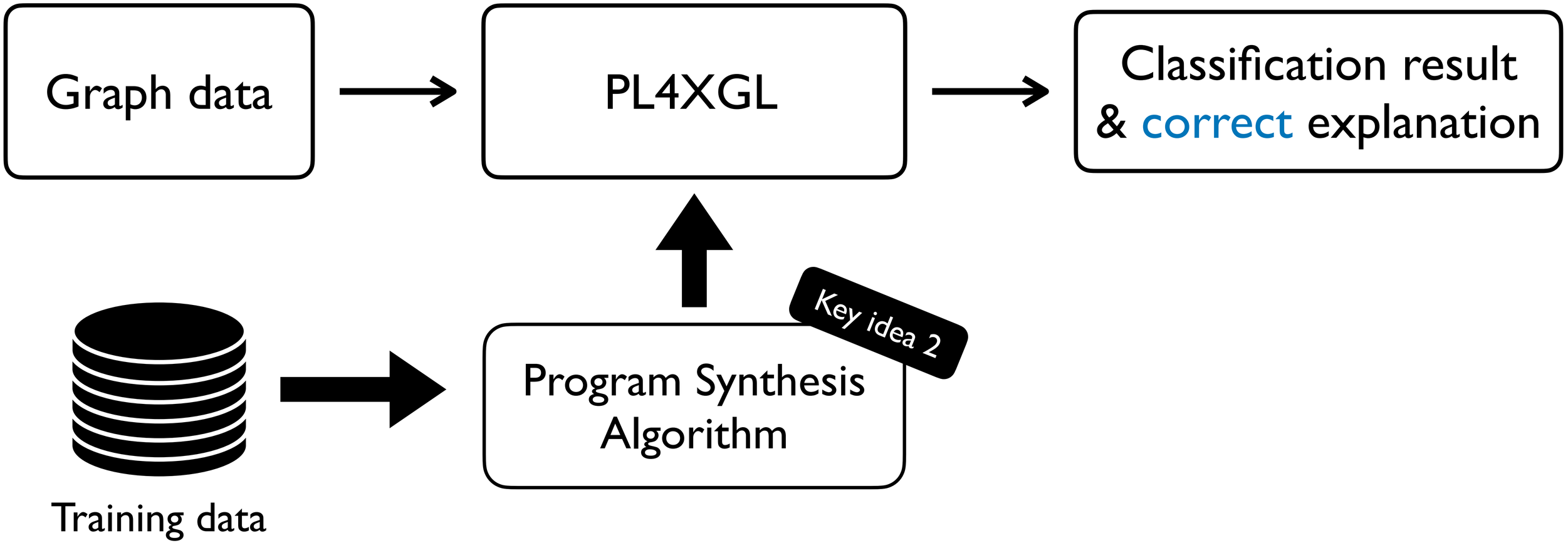
Our Approach

- PL4XGL: PL-based inherently explainable graph machine learning method

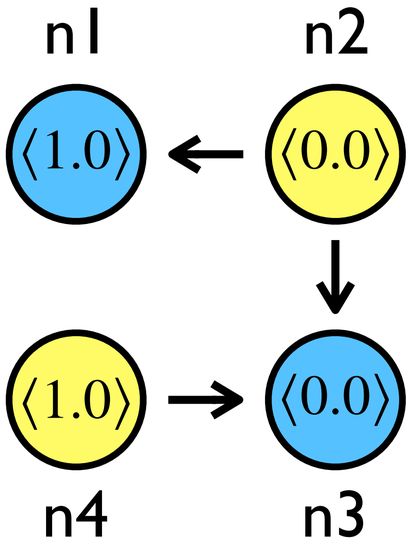


Our Approach

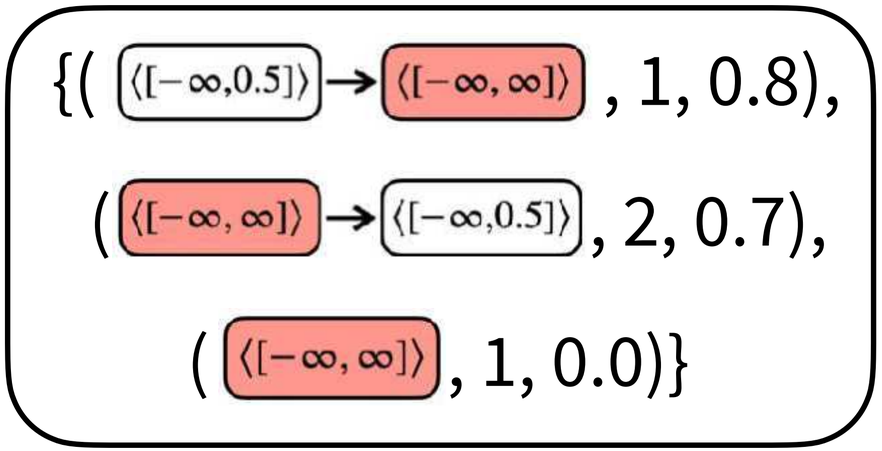
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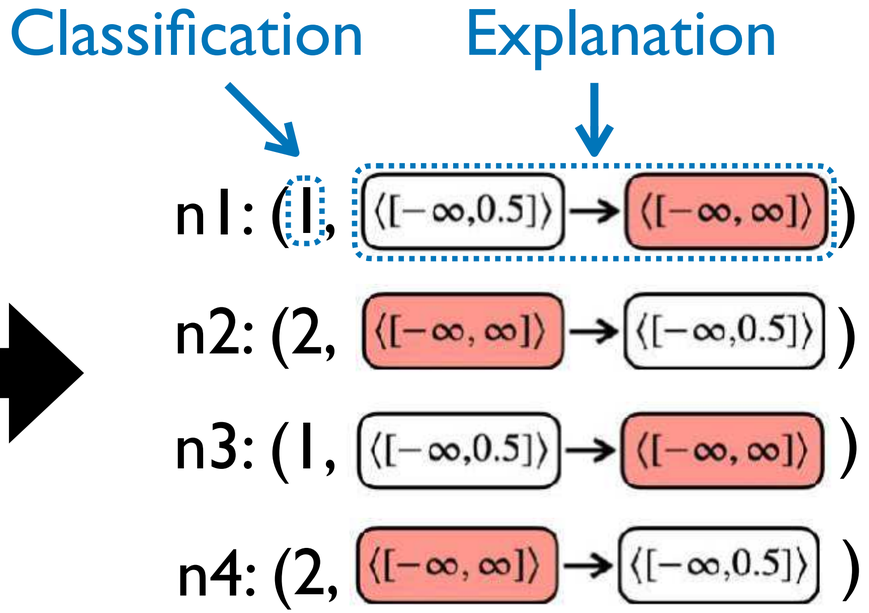
Node Classification Example



Graph data

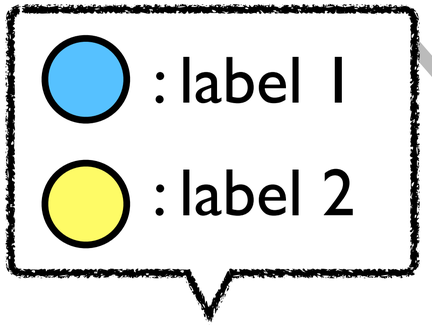


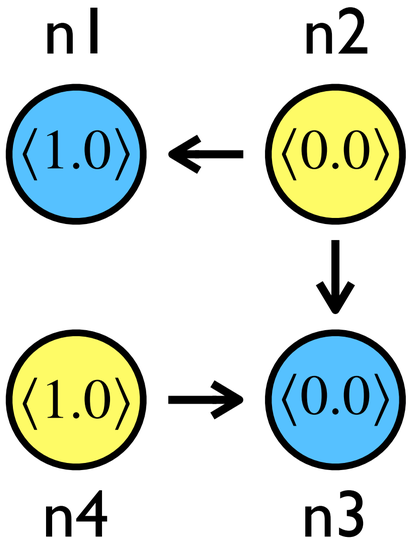
Our model



Classification & Explanation

Node Classification Example





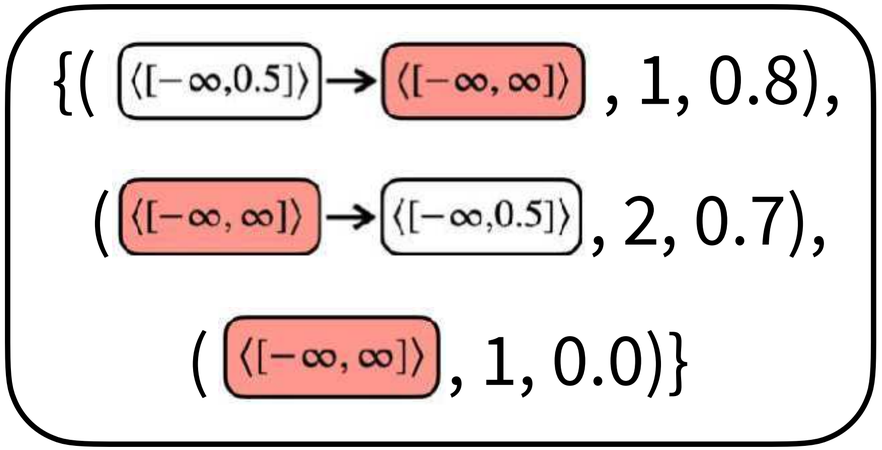
Graph data

Our model

Classification & Explanation

Node Classification Example

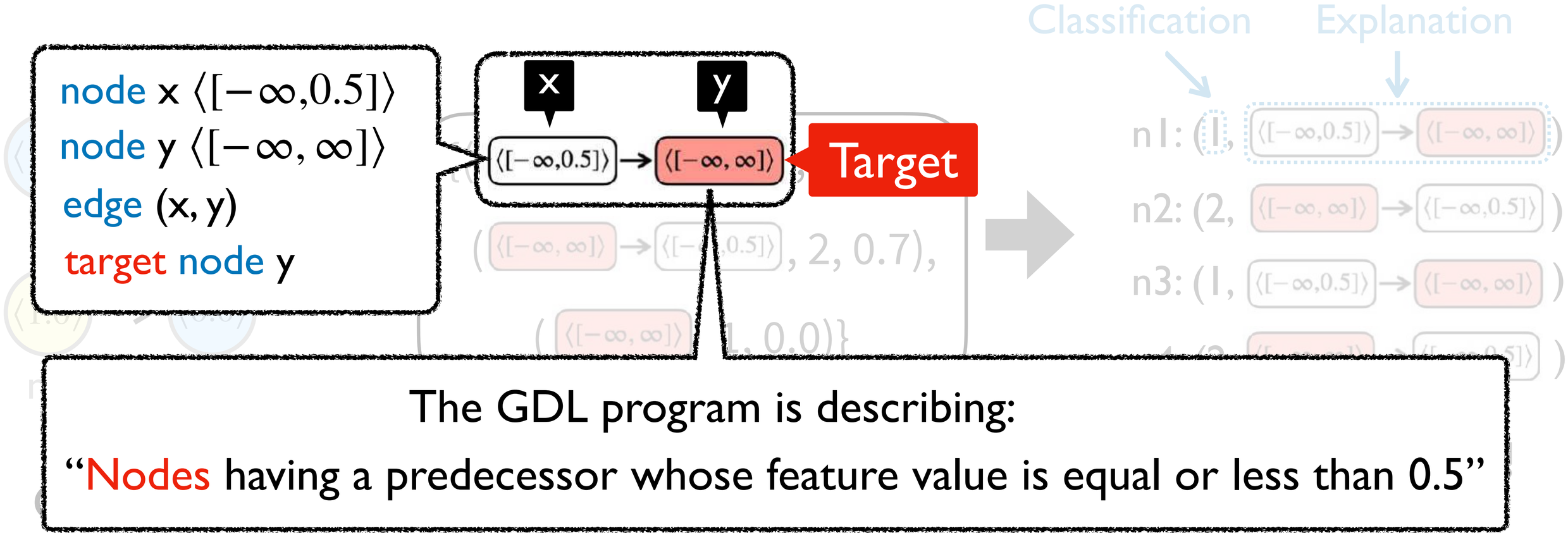
Graph data



Our model

Classification & Explanation

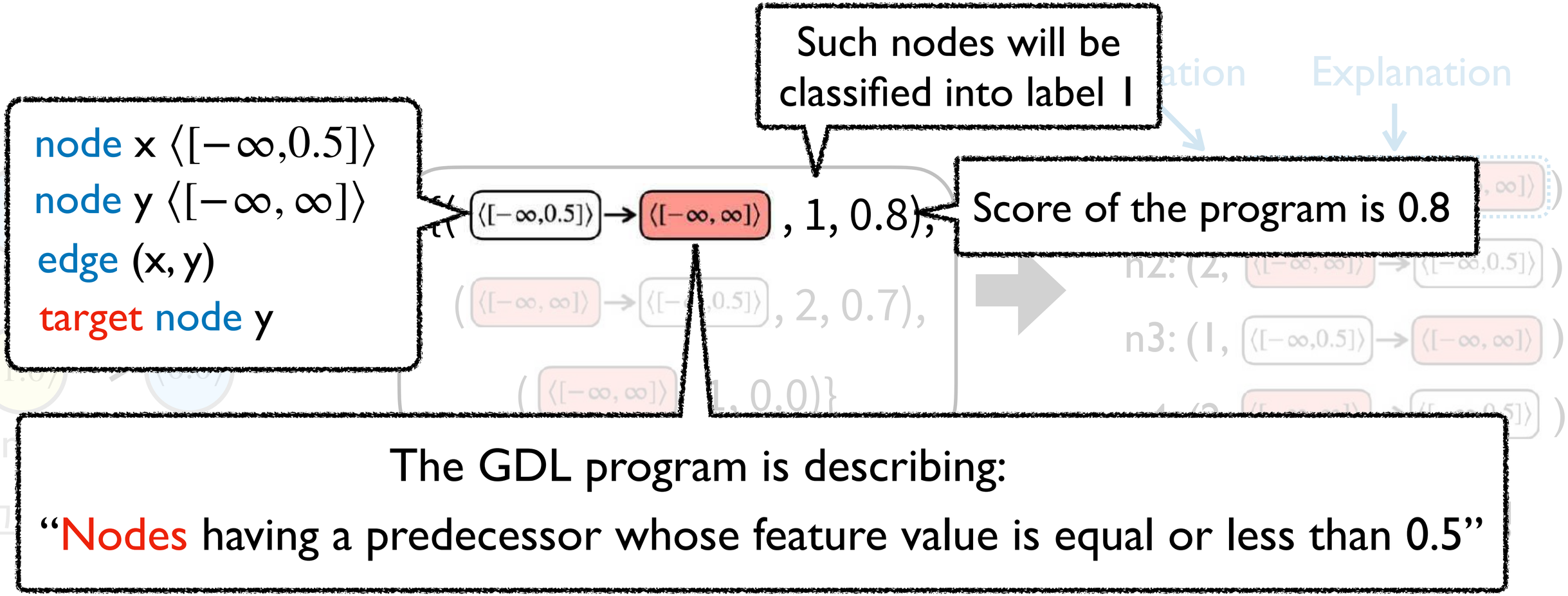
Node Classification Example



The GDL program is describing:

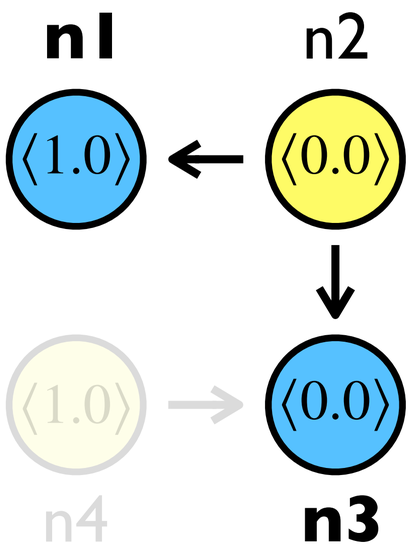
"Nodes having a predecessor whose feature value is equal or less than 0.5"

Node Classification Example

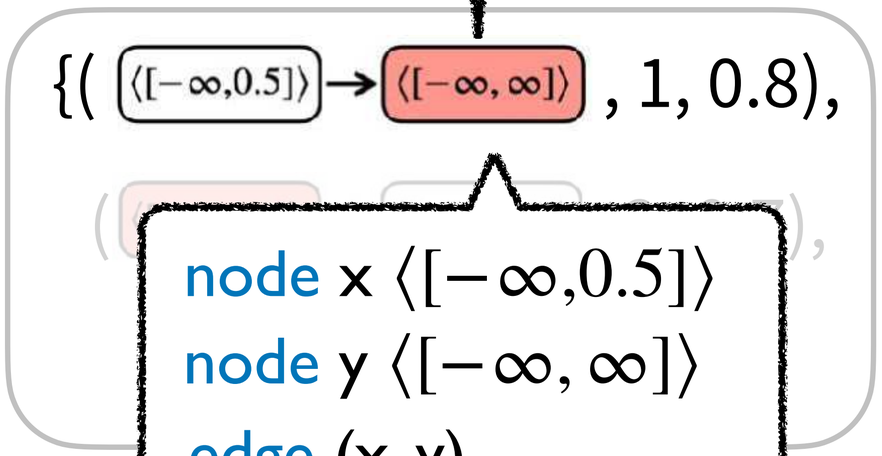


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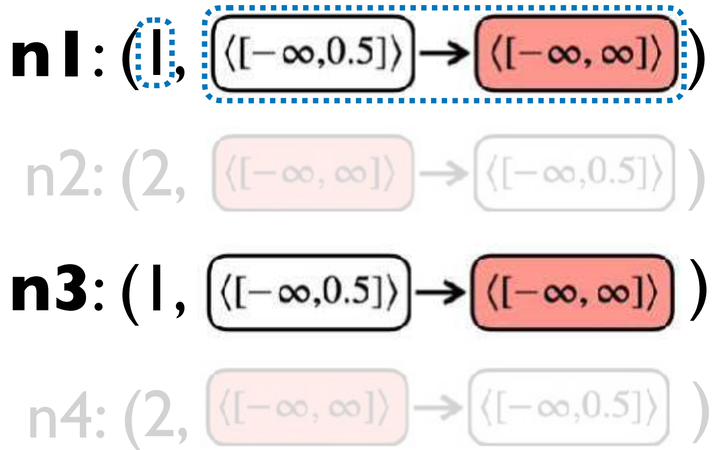


Graph data



Our model

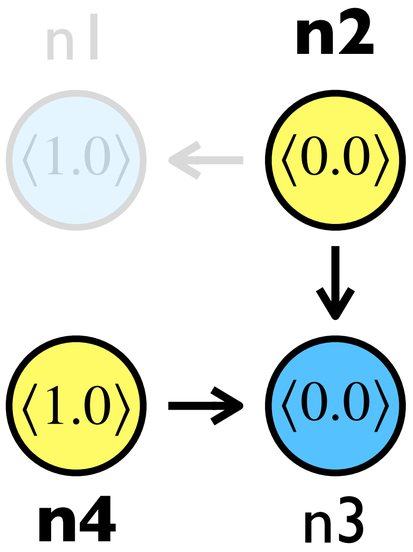
Classification Explanation



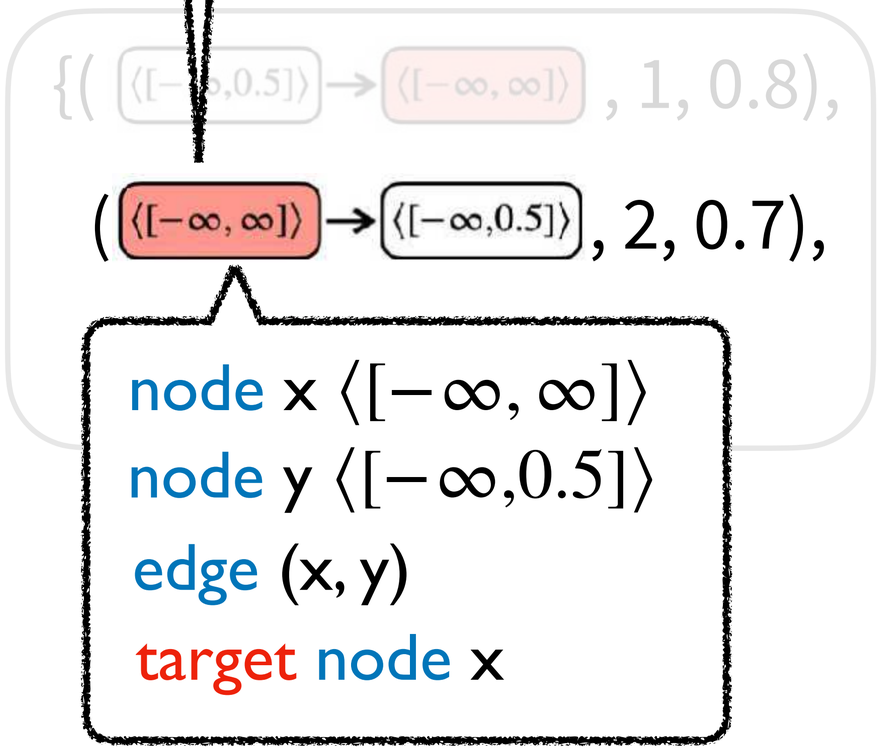
Classification & Explanation

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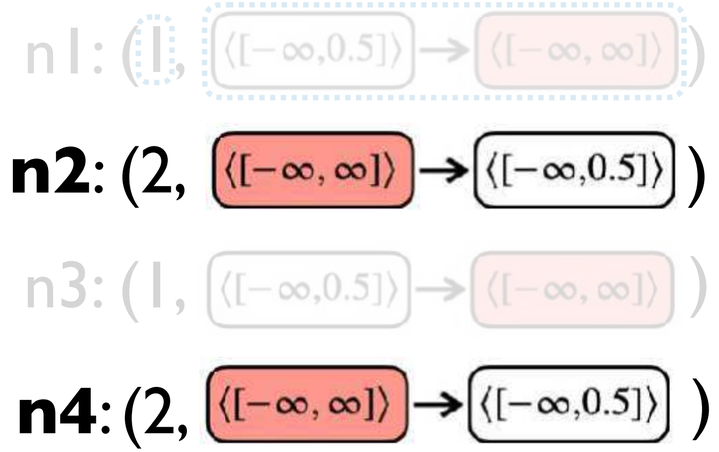
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Graph data

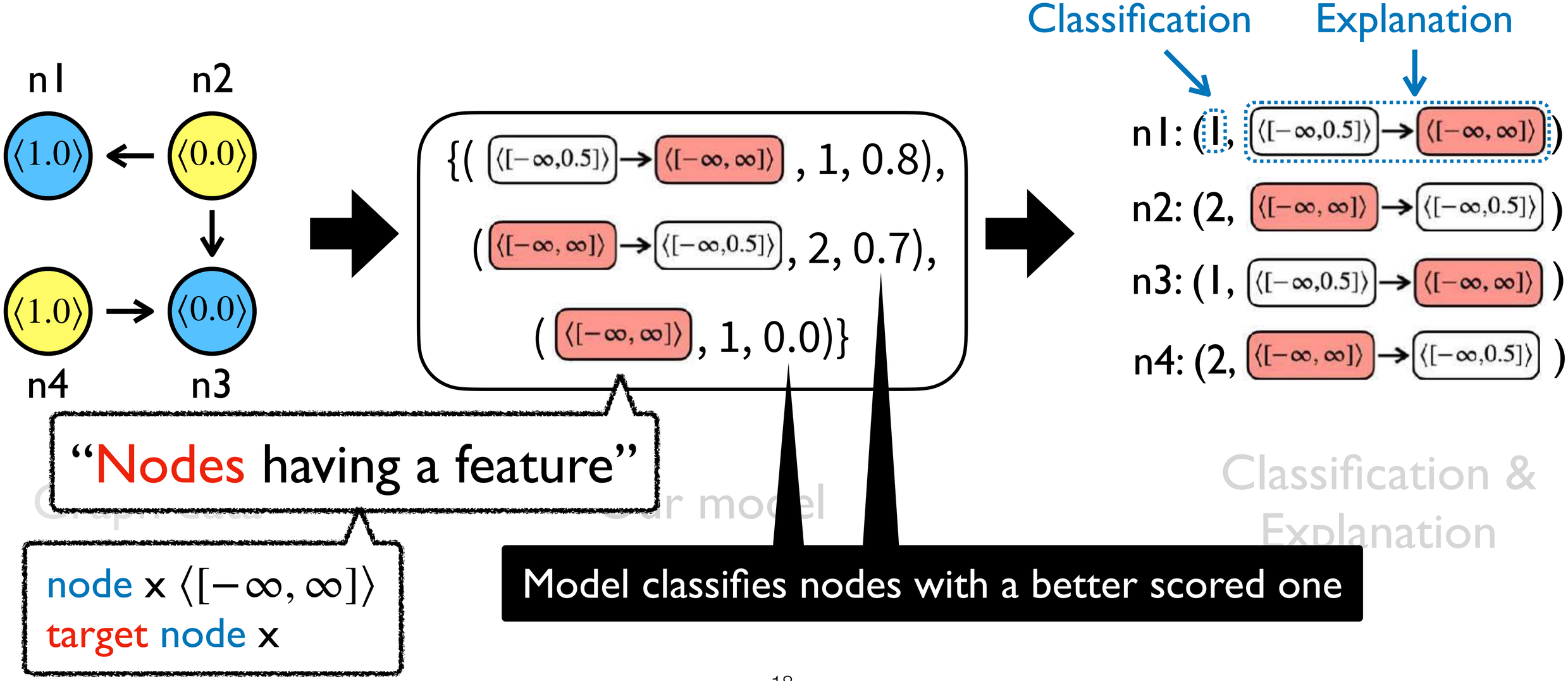


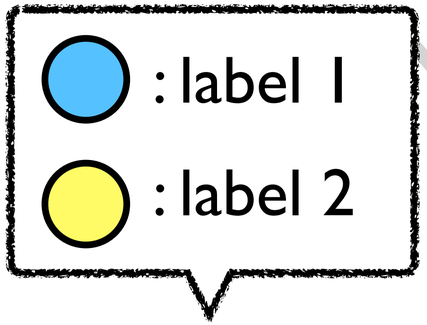
Classification Explanation

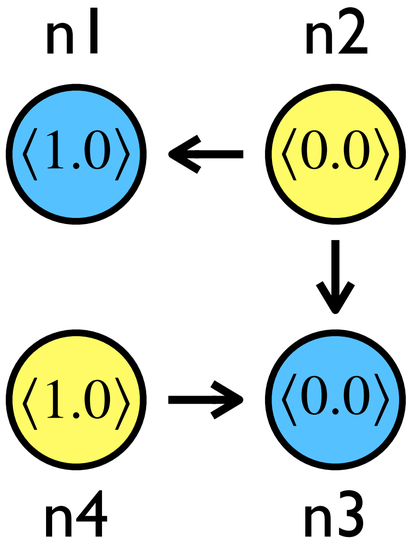


Classification & Explanation

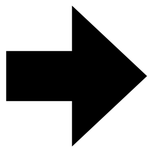
Node Classification Example

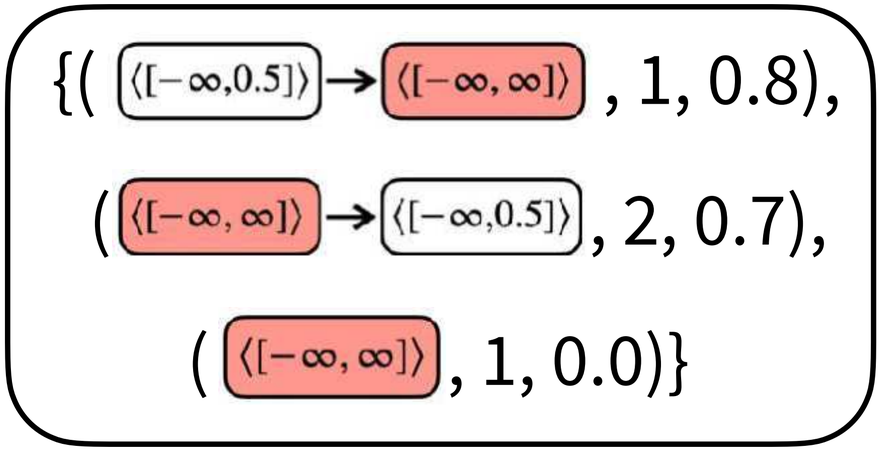




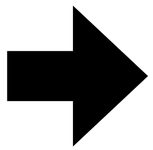


Graph data



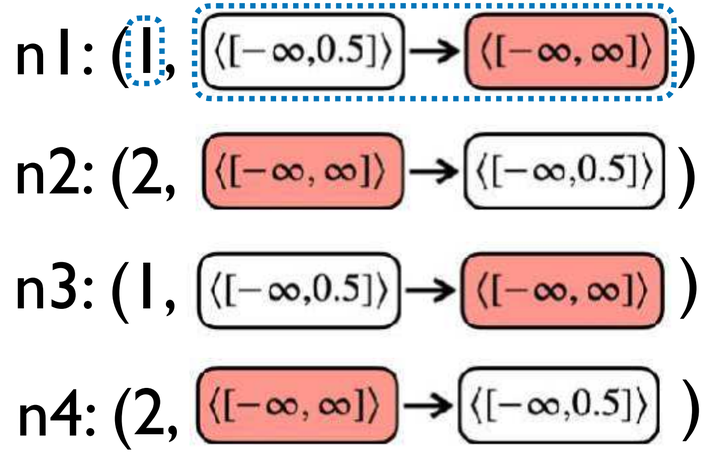


Our model



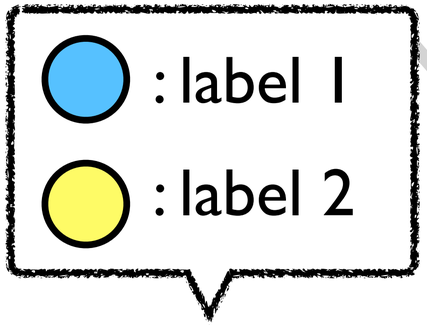
- No additional explanation cost
- Explanations are guaranteed to be correct

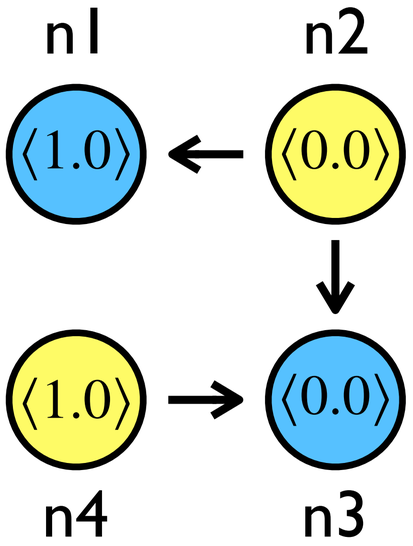
Classification Explanation



Classification & Explanation

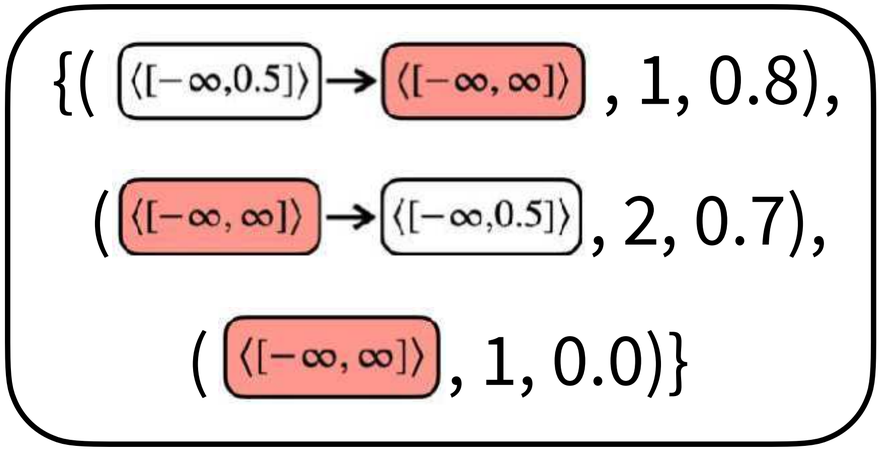
Node Classification Example





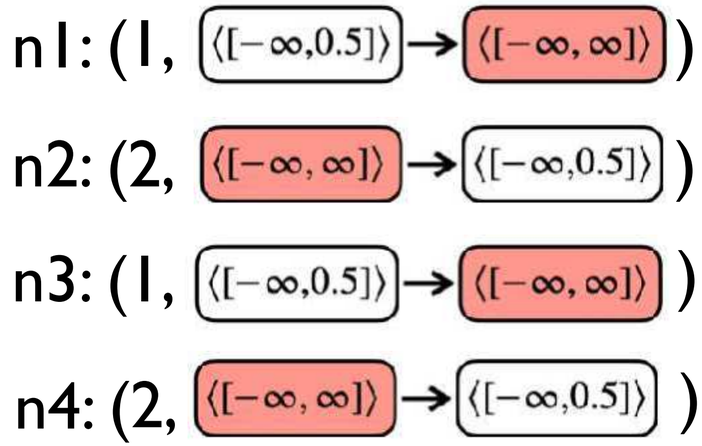
Graph data

Quality of the programs determines the accuracy



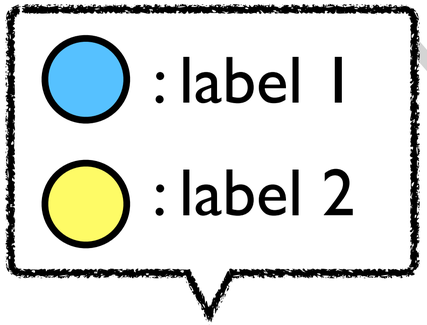
Our model

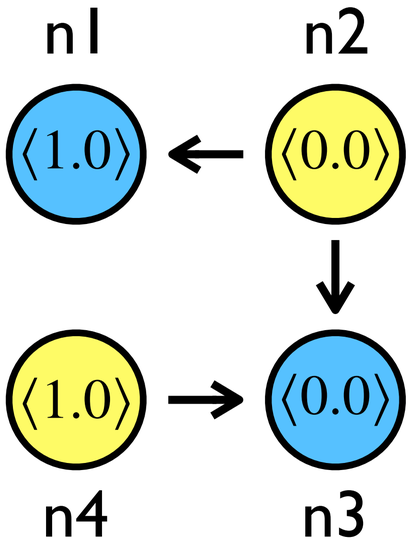
Accuracy : **1.0**



Classification & Explanation

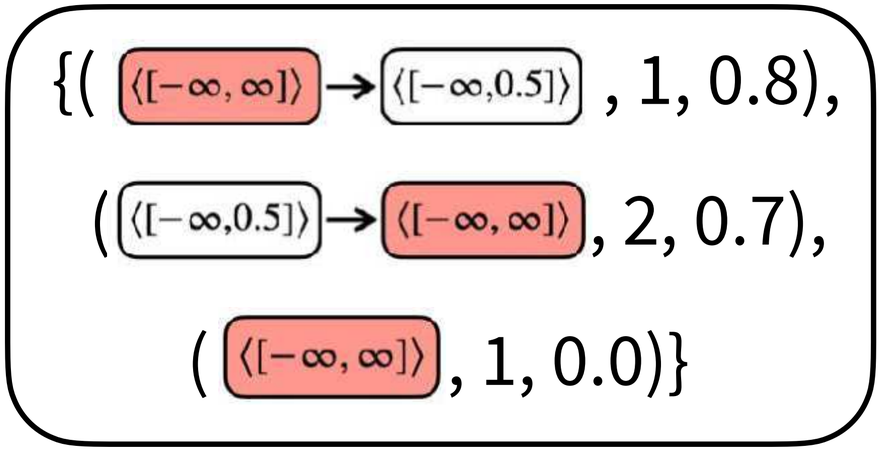
Node Classification Example





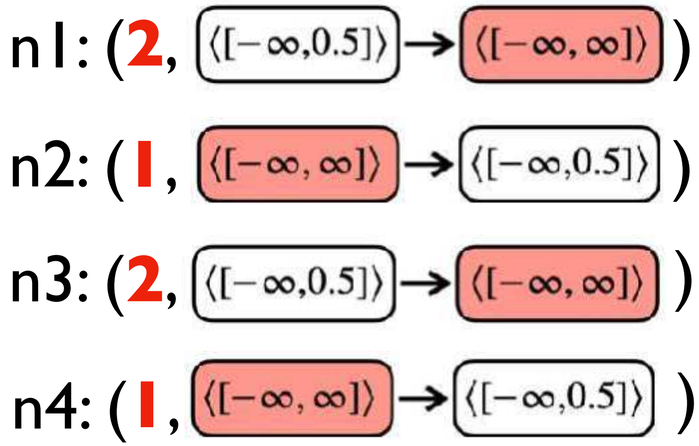
Graph data

Quality of the programs determines the accuracy



Our model

Accuracy : **0.0**



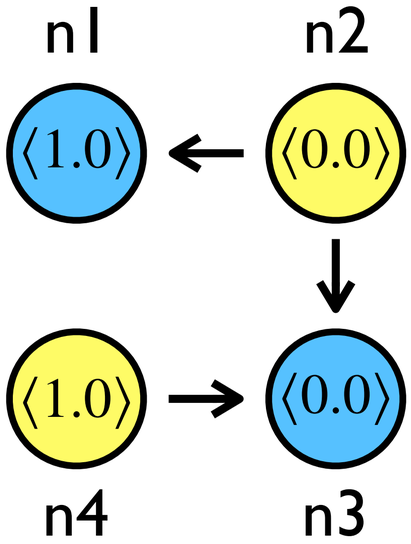
Classification & Explanation



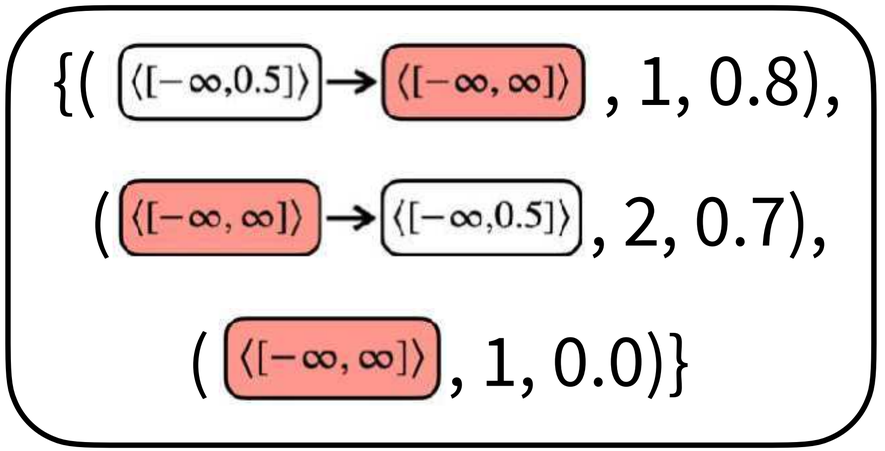
Training data

Learning algorithm

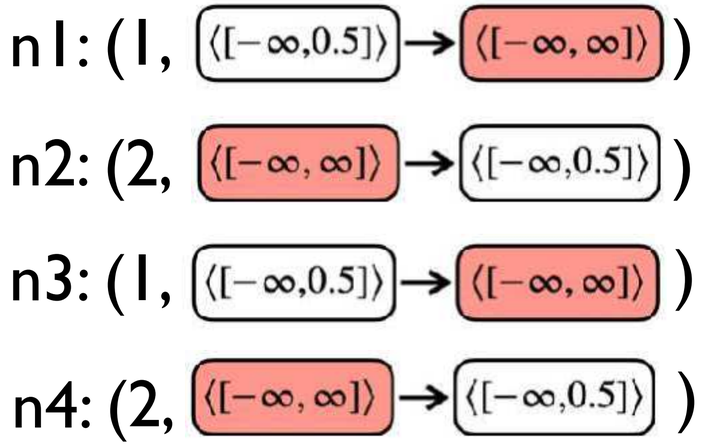
Learning objective:
Learn high-quality GDL programs



Graph data



Our model



Classification & Explanation



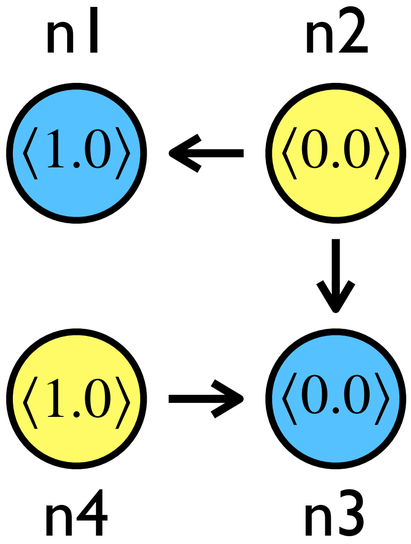
Training data

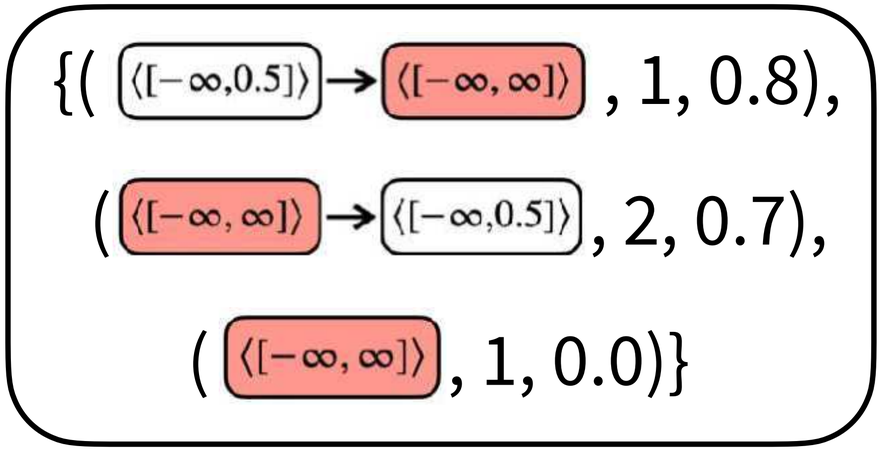
Top-down synthesis algorithm

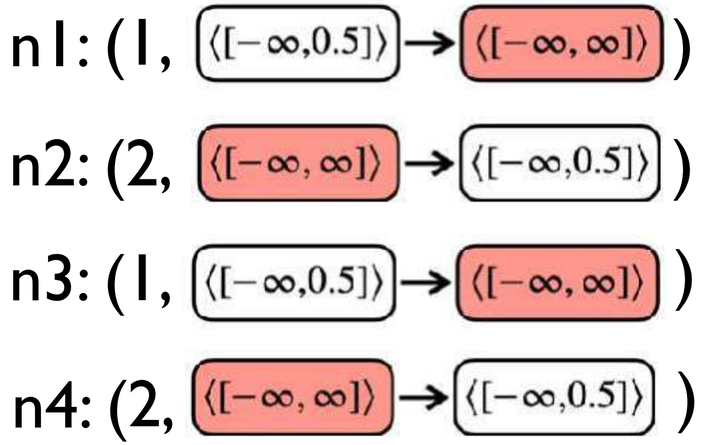
Bottom-up synthesis algorithm

Learning algorithm

Learning objective:
Learn high-quality GDL programs







Graph data

Our model

Classification & Explanation

Evaluation

- Compared PL4XGL with
  - Representative GNNs : GCN, GAT, GIN, etc
  - State-of-the-art GNN explainer : SubgraphX*
- Research questions:
  - RQ1) Classification accuracy
  - RQ2) Explainability
- Settings:
  - GNNs and SubgraphX trained and evaluated using a GPU (RTX A6000)
  - PL4XGL trained and evaluated using 64-core CPU

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*Yuan et al. On explainability of graph neural networks via subgraph explorations. ICML 2021

RQ1) Classification Accuracy

- Each dataset is split into 8:1:1 for training, validation, and evaluation
- PL4XGL achieved the best accuracy for 5 datasets
- PL4XGL did not scale for the largest dataset HIV (time budget = 48h)

	GCN	GAT	ChebyNet	JKNet	GraphSage	GIN	DGCN	PL4XGL
MUTAG	80.0±0.0	89.0±2.2	86.0±4.1	68.0±7.5	78.0±4.4	91.0±5.4	N/A	**100.0±0.0**
BBBP	83.6±1.4	82.3±1.6	84.6±1.0	85.6±1.9	86.6±0.9	86.2±1.4	N/A	**86.8±0.0**
BACE	78.4±2.8	52.4±3.3	78.9±1.4	79.9±1.9	79.8±0.8	**80.9±0.4**	N/A	**80.9±0.0**
HIV	96.4±0.0	96.4±0.0	96.8±0.2	96.8±0.1	**96.9±0.2**	96.8±0.1	N/A	N/A
BA-Shapes	95.1±0.6	76.8±2.3	**97.1±0.0**	94.3±0.0	**97.1±0.0**	92.0±1.1	95.1±0.7	95.7±0.0
Tree-Cycles	97.7±0.0	90.9±0.0	**100.0±0.0**	98.9±0.0	**100.0±0.0**	93.2±0.0	99.2±0.5	**100.0±0.0**
Wisconsin	64.0±0.0	49.6±3.1	86.4±3.9	64.8±1.5	92.8±2.9	56.0±0.0	**96.0±0.0**	88.0±0.0
Texas	67.7±5.3	50.0±0.0	87.7±2.1	68.8±4.3	86.6±2.6	50.0±0.0	**86.6±2.6**	83.3±0.0
Cornell	58.9±2.6	61.1±0.0	81.0±6.5	61.1±0.0	87.7±2.1	61.1±0.0	86.6±2.6	**88.8±0.0**
Cora	85.6±0.3	86.4±1.8	86.5±5.2	84.9±3.5	86.3±3.2	**86.7±0.0**	83.2±5.9	80.0± 0.0
Citeseer	75.2±0.0	74.3±0.7	**79.1±0.9**	73.7±4.2	75.9±2.3	75.2±0.0	71.3±6.0	63.8± 0.0
Pubmed	82.8±1.1	84.7±1.2	**88.7±1.0**	83.2±0.4	88.0±0.4	86.1±0.6	85.1±0.6	81.4±0.0

RQ1) Classification Accuracy

- Each dataset is split into 8:1:1 for training, validation, and evaluation
- PL4XGL
- PL4XGL did not scale for the largest dataset HIV (time budget = 48h)

Molecule datasets (graph classification)

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BBBP	83.6±1.4	82.3±1.6	84.6±1.0	85.6±1.9	86.6±0.9	86.2±1.4	N/A	**86.8±0.0**
BACE	78.4±2.8	52.4±3.3	78.9±1.4	79.9±1.9	79.8±0.8	**80.9±0.4**	N/A	**80.9±0.0**
HIV	96.4±0.0	96.4±0.0	96.8±0.2	96.8±0.1	**96.9±0.2**	96.8±0.1	N/A	N/A
BA-Shapes	95.1±0.6	76.8±2.3						
Tree-Cycles	97.7±0.0	90.9±0.0						
Wisconsin	64.0±0.0	49.6±3.1						
Texas	67.7±5.3	50.0±0.0						
Cornell	58.9±2.6	61.1±0.0	81.0±6.5	61.1±0.0	87.7±2.1	61.1±0.0	86.6±2.6	**88.8±0.0**
Cora	85.6±0.3	86.4±1.8	86.5±5.2	84.9±3.5	86.3±3.2	**86.7±0.0**	83.2±5.9	80.0± 0.0
Citeseer	75.2±0.0	74.3±0.7	**79.1±0.9**	73.7±4.2	75.9±2.3	75.2±0.0	71.3±6.0	63.8± 0.0
Pubmed	82.8±1.1	84.7±1.2	**88.7±1.0**	83.2±0.4	88.0±0.4	86.1±0.6	85.1±0.6	81.4±0.0

PL4XGL shows the best accuracy

RQ1) Classification Accuracy

- Each dataset is split into 8:1:1 for training, validation, and evaluation
- PL4XGL achieved the best accuracy for 5 datasets
- PL4XGL did not scale for the largest dataset HIV (time budget = 48h)

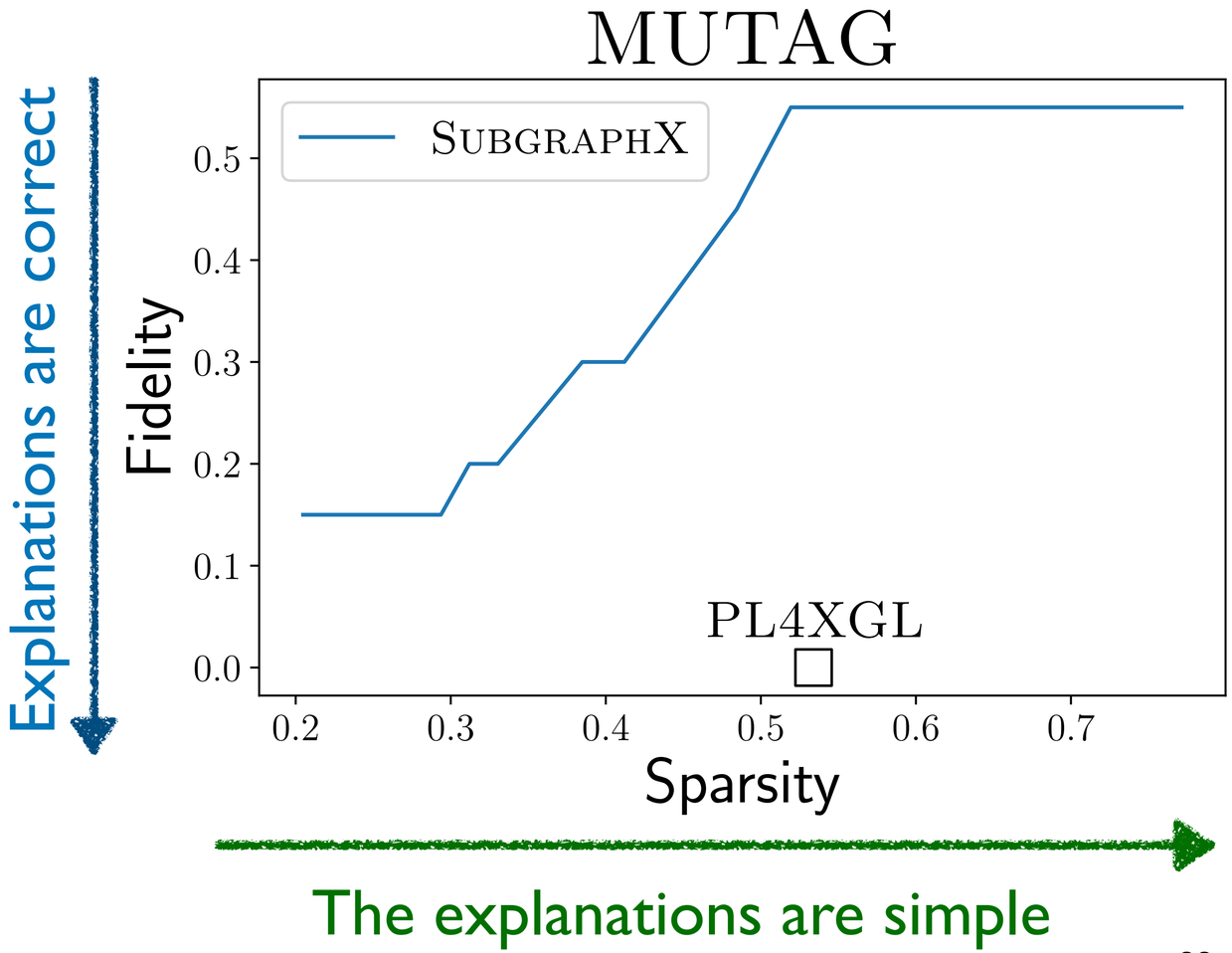
	GCN	GAT	ChebyNet	JKNet	GraphSage	GIN	DGCN	PL4XGL
MUTAG	80.0±0.0	89.0±2.2	86.0±4.1	68.0±7.5	78.0±4.4	91.0±5.4	N/A	**100.0±0.0**
BBBP	83.6±1.4	82.3±1.6	84.6±1.0	85.6±1.9	86.6±0.9	86.2±1.4	N/A	**86.8±0.0**
BACE	78.4±2.8	52.4±3.3	78.9±1.4	79.9±1.9	79.8±0.8	**80.9±0.4**	N/A	**80.9±0.0**
HIV	96.4±0.0	96.4±0.0	96.8±0.2	96.8±0.1	**96.9±0.2**	96.8±0.1	N/A	N/A
BA-Shapes	95.1±0.6	76.8±2.3	**97.1±0.0**	94.3±0.0	**97.1±0.0**	92.0±1.1	95.1±0.7	95.7±0.0
Pubmed	82.8±1.1	84.7±1.2	**88.7±1.0**	83.2±0.4	88.0±0.4	86.1±0.6	85.1±0.6	81.4±0.0

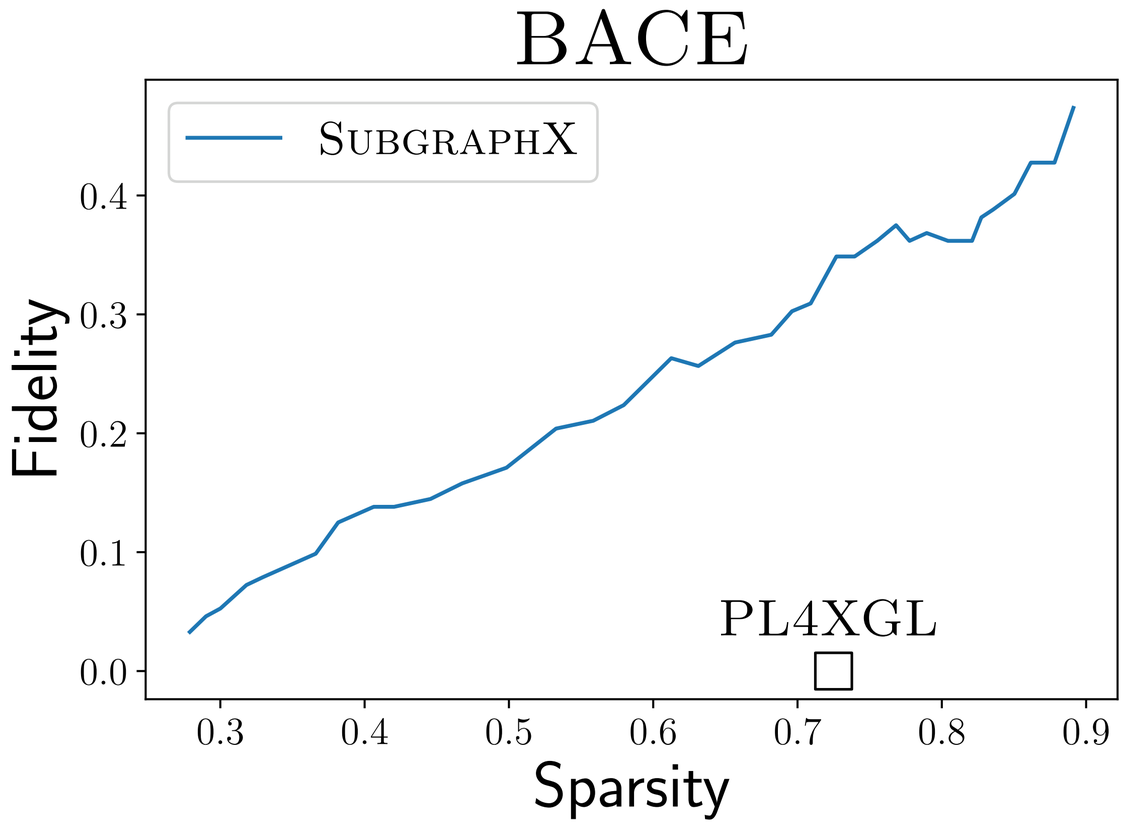
PL4XGL failed its training in HIV dataset because of its training cost

- HIV includes 41,127 (1,049,163 nodes)
- Timeout = 2 day (48 hours)

RQ2) Explainability

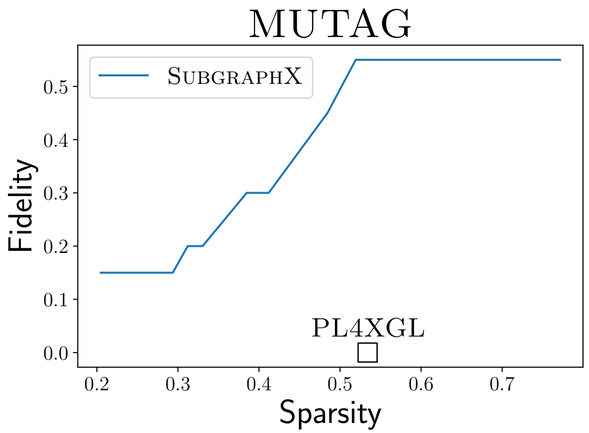
- Our approach provides correct & simple explanations

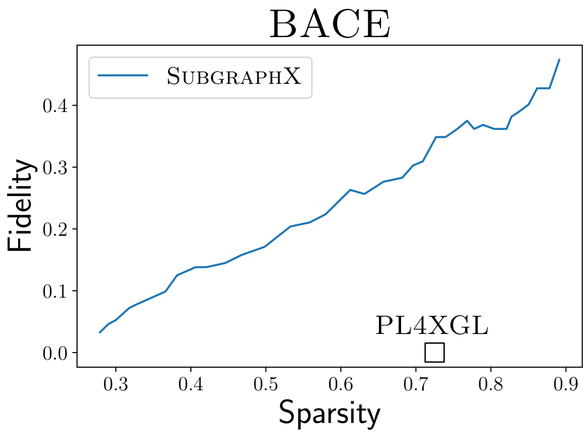


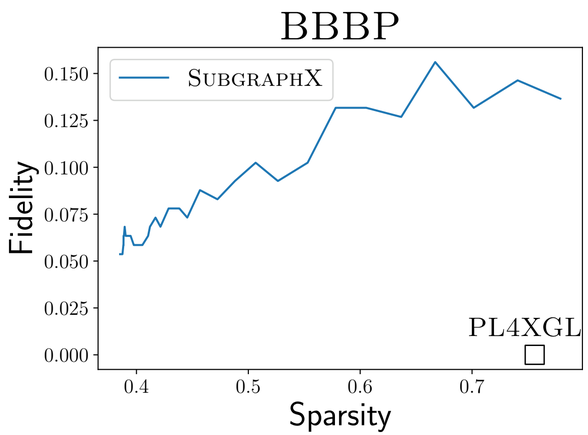


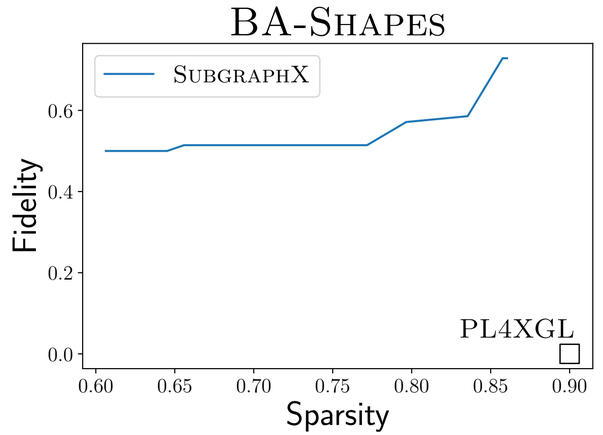
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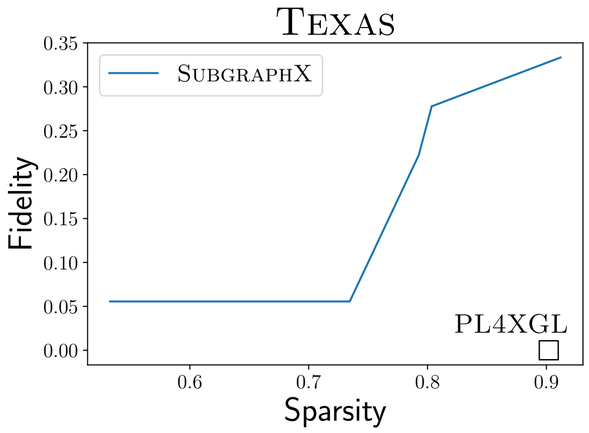
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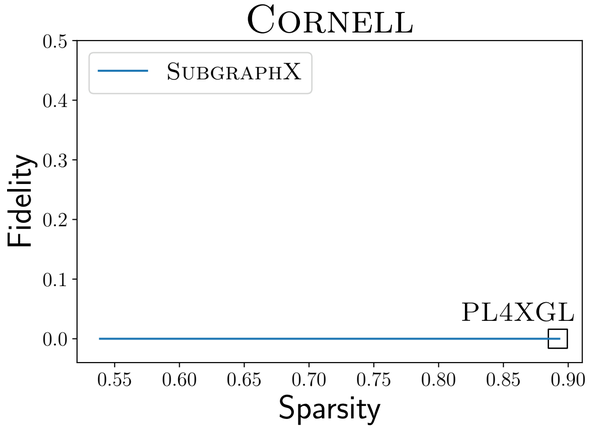


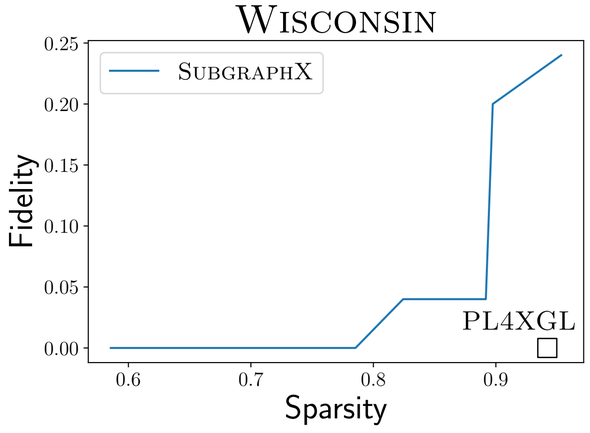


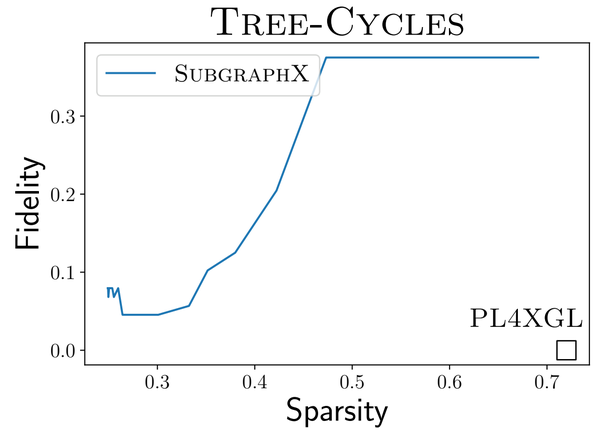












Summary

- Problem : Accurate and explainable graph learning
- Solution : A purely PL-based approach to XAI
  - Domain specific language design for defining AI models
  - Program synthesis for learning models from training data
- Result:
  - Accuracy can compete with GNNs
  - Better explainability than GNNs with post-hoc explainer

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Conclusion: PL techniques are even useful for AI!